Progress Report 4 Video Outline

Gabriel Anokye | Senior Seminar | Target duration 4 minutes 45 seconds

Record the Waypoint extension beside the fixture page at 1080p or higher. Use Chrome for Testing, which exposes the debugging port. Open the side panel and increase browser zoom enough that the run details remain readable.

Before recording

- Build the latest extension and runner with pnpm build. Start the fixture server with pnpm serve:mutations.
- Start the runner with BROWSER_HEADLESS=false BROWSER_SLOW_MO=200 pnpm --filter @waypoint/runner start.
- Start pnpm chrome:debug "http://127.0.0.1:8080/forms.html?mutation=attribute&seed=demo" and open Waypoint in Chrome for Testing.
- Have the report PDF open on its results table. Check that DEMO — automatic repair (Sign in) and LIVE — your recording on a mutated page are listed. If absent, use the controlled screenshots in the report for those scenes.
- Before each attached run, check Use selected page for the starting URL. Keep Reload selected tab unchecked. Refresh the fixture between comparisons.
- Do not wait for the full N=5 benchmark on camera; show the recorded output and source artifacts. This outline attributes the results to the completed 34,440-trial run.

Timed scenes and narration

0:00 to 0:30 Research question

ON SCREEN Show the report title, then the forms fixture.

SAY “I am Gabriel Anokye. My research asks how reliably browser automations find the same target after a page changes, and whether an algorithm can repair a broken step without clicking the wrong control.”

0:30 to 1:05 Corpus and method

ON SCREEN Show the report mutation table and the fixture mutation URL. Point to the 16 page states and 177 truth declarations.

SAY “I finished six mutation classes across eight fixture pages and eight poorly labeled variants. The ground-truth ID stays fixed while layout, structure, content, attributes or semantics change. The audit preserved all 2,478 compared anchor occurrences.”

1:05 to 1:45 Healing off

ON SCREEN Show forms.html?mutation=attribute&seed=demo. In Replay options, leave healing off; run DEMO — automatic repair (Sign in), then open the failed run.

SAY “The recorded test ID was renamed by the attribute mutation. With healing off, Waypoint reaches the wait step, cannot uniquely locate the button and stops. The target action is not executed.”

1:45 to 2:35 Healing on

ON SCREEN Refresh the same URL, check Enable self-healing and run the same DEMO workflow. Expand the click result showing Correct target, score and margin.

SAY “With healing on, Waypoint compares visible candidates with the recorded element snapshot. Sign in scores 1.0000 and leads the next candidate by 0.5633, clearing both gates. The actual trusted click reaches login-submit. The form can still ask for required fields; the measured action here is the correct click.”

2:35 to 3:25 Safe refusal

ON SCREEN Open ambiguous.html?mutation=structural&seed=demo. Use LIVE — your recording on a mutated page, or the latest ambiguous recording. Keep both replay switches on. Open the manual-intervention run.

SAY “The three Delete controls have nearly identical names. Here the highest score is about 0.9583, but its lead is only 0.0154, below the 0.15 safety margin. Waypoint stops at waitFor before any Delete click. The ranked list is diagnostic; choosing a control requires Rebind element in the editor and a new run.”

3:25 to 4:15 Measured results

ON SCREEN Show report summary and threshold table. Point to Pass A 66.31%, Pass B 81.84%, 2,609 recoveries and 20 greedy wrong Delete clicks.

SAY “Across 16,800 action evaluations per pass, correct-target rate increased by 15.53 percentage points. Of 5,524 broken actions, 2,609 were repaired correctly, none were repaired to a wrong target, 2,390 were safely escalated and 525 remained unresolved. In a 210-trial forced-broken Delete sweep, greedy top one made 20 wrong clicks; every gated setting refused all 210.”

4:15 to 4:45 Scope and next step

ON SCREEN Show the report limitations paragraph and raw CSV/manifest files.

SAY “These are independent actions on a fixed synthetic corpus, not complete website workflows or independent website samples. The remaining 136 wrong actions came from baseline locators that resolved incorrectly before healing could run. The next evaluation should use unfamiliar sites and measure human repair time.”

Recording cues

- Keep the address bar visible when changing mutation URLs so the viewer can see the condition.
- Pause on the expanded Step results long enough to read the score, margin, expected truth ID and actual truth ID.
- Show that no Delete status message appears in the browser after manual intervention. The run pauses at an inferred wait step before a click.
- If a live run varies, use Figures 2 to 4 and the evidence JSON from the controlled local session; describe them as recorded evidence, not as a fresh live attempt.
- The benchmark takes about twelve minutes on the measured machine. Use the completed run output or the report during the video, and say it is the recorded N=5 result.